

IoT Based Healthcare System

Submitted in partial fulfillment of the requirements for the course

Microprocessors and Microcontrollers (BECE204L)

by

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1. INTRODUCTION

1.1. Objective

To design and implement an IoT-based healthcare system for the real-time monitoring of patients' vital signs such as heart rate, blood pressure, and temperature using interconnected sensors. This system provides remote monitoring capabilities, reducing the need for frequent hospital visits, improving early detection of health issues, and enhancing the overall efficiency of Healthcare delivery.

1.2. Problem Statement

Worldwide Healthcare systems are facing challenges due to the overcrowding of hospitals, leading to long waiting times, reduced access to quality healthcare and an increased burden on patients with chronic diseases. Monitoring patients' vital signs, such as heart rate, blood pressure, and body temperature, typically requires regular in-person visits, which not only increases the strain on healthcare facilities but also limits the ability to track patients' health continuously.

Traditional methods of healthcare often lead to delayed diagnosis, unnecessary hospital visits, and an increased cost. Patients with chronic illnesses require continuous monitoring of their vital signs which is difficult to achieve without continuous real-time tracking.

These problems can be solved by an IoT-based healthcare system that makes it possible to continuously monitor vital signs remotely, detect health problems early, and enable quick action from medical personnel. In addition to lessening the strain on medical facilities, this will increase access to high-quality care, particularly for patients in rural and isolated locations, improving healthcare delivery as a whole.

This problem is directly aligned with several **UN Sustainable Development Goals (SDGs)**, particularly:

1) SDG 3: Good Health and Well-being

The proposed system increases access to quality healthcare leading to better healthcare outcomes especially in low income regions. Remote monitoring also ensures timely

medical care for patients with chronic conditions.

2) SDG 9: Industry, Innovation, and Infrastructure

The IoT-based solution promotes innovation in healthcare delivery. The system relies on modern technologies like IoT, cloud computing to create a more efficient and resilient healthcare infrastructure.

3) SDG 10: Reduced Inequality

The remote monitoring capability of the system is beneficial for patients in rural and marginalised communities. It ensures that healthcare is more inclusive and accessible regardless of geographic location.

1.3. Key contributions

- Multi-Parameter Monitoring – Combines MAX30100 (HR & SpO2), AD8232 (ECG), and DS18B20 (Temp) for real-time evaluation of health metrics.
- Wireless Data & Remote Access – Employs ESP32 Wi-Fi to transmit vital signs to ThingSpeak for off-site monitoring.
- Enhanced ECG Accuracy – Signal amplification and noise reduction guarantee accurate ECG measurements.
- Portable & Low-Power Design – Designed for ongoing, mobile assessment with minimal power consumption.
- Customizable Alerts – Issues notifications via ThingSpeak when vital signs surpass established thresholds for prompt action.
- Real-Time Data Visualization – Live health metrics can be accessed from anywhere through the cloud.
- Seamless IoT Integration – Utilizes the ThingSpeak API for effortless connectivity within healthcare systems.
- Improved Patient Safety – Facilitates early detection of anomalies and prompt interventions.

2. SYSTEM MODEL

2.1 Proposed Methodology Design/Approach

The creation of this IoT-oriented healthcare monitoring system follows a methodical framework to guarantee precision, dependability, and ease of use. The design process initiates with establishing user and data specifications, concentrating on the ongoing monitoring of vital signs with a high level of precision. The choice of appropriate hardware, which includes the ESP32 microcontroller and biomedical sensors, guarantees smooth integration and compatibility. Data acquisition from sensors is realized through effective coding methods to ensure real-time processing with reduced latency. Sophisticated signal processing methods are utilized on ECG data to minimize noise and deliver precise heart activity assessment. Wireless communication is facilitated via the ESP32's integrated Wi-Fi, allowing the transmission of real-time data to ThingSpeak for remote access and analysis. A user-friendly OLED interface is designed to display vital signs and alerts locally, improving usability. Power management strategies are optimized to prolong battery life, making the system suitable for uninterrupted monitoring. Thorough testing and validation occur under diverse circumstances to ensure precision, followed by deployment strategies that include user training and considerations for data security to provide a reliable and scalable solution.

- **Define User & Data Requirements:** Determine target users, the environment for monitoring, and necessary accuracy.
- **Select Suitable Hardware:** Opt for the ESP32, MAX30100, AD8232, DS18B20, and OLED display to achieve seamless integration.
- **Develop Sensor Data Acquisition:** Execute accurate data collection and processing algorithms.
- **Implement ECG Signal Processing:** Execute noise reduction and amplification techniques for precise ECG measurements.
- **Enable Wireless Data Transmission:** Leverage the ESP32's Wi-Fi to send real-time data to ThingSpeak.
- **Design User-Friendly OLED Interface:** Create an intuitive display for real-time information and alerts.
- **Configure ThingSpeak for Data & Alerts:** Establish cloud storage, data visualization, and automated notifications.

- Optimize Power Management: Apply energy-saving strategies for continuous operation.
- Conduct Rigorous Testing & Validation: Calibrate sensors and evaluate the system under varied conditions.
- Plan for Deployment & Support: Offer user training, options for remote monitoring, and measures for data security.

2.2 Algorithm/code

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include "MAX30100_PulseOximeter.h"
//#include ""

const char* ssid = "Akash's Galaxy A54 5G";
const char* password = "Akash Nidagundi";

String apiKey = "RLSMSS3141KUWAA0";
String server = "http://api.thingspeak.com/update";

#define OLED_SDA 21
#define OLED_SCL 22
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

#define ECG_PIN 34
#define TEMP_SENSOR 35
```

```

#define BUZZER_PIN 25
#define LED_PIN 26

OneWire oneWire(TEMP_SENSOR);
DallasTemperature sensors(&oneWire);

PulseOximeter pox;

unsigned long lastMedicineTime = 0;
const unsigned long reminderInterval = 4 * 60 * 60 * 1000;
const int ECG_THRESHOLD = 2000; // Adjust this threshold based on your needs

String getHeartRateStatus(float hr) {
    if (hr == 0 || isnan(hr)) return "No Finger Detected";
    if (hr < 50) return "Low HR";
    if (hr > 120) return "High HR";
    return "Normal";
}

String getTimeLeft(unsigned long lastTime, unsigned long interval) {
    unsigned long elapsed = millis() - lastTime;
    unsigned long timeLeft = interval - elapsed;

    if (timeLeft <= 0) return "00:00";

    int hours = timeLeft / (60 * 60 * 1000);
    int minutes = (timeLeft % (60 * 60 * 1000)) / (60 * 1000);

    return (hours < 10 ? "0" : "") + String(hours) + ":" +
        (minutes < 10 ? "0" : "") + String(minutes);
}

void setup() {
    Serial.begin(115200);

```

```

pinMode(BUZZER_PIN, OUTPUT);
pinMode(LED_PIN, OUTPUT);
pinMode(ECG_PIN, INPUT);

Wire.begin();

if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("SSD1306 allocation failed");
    while (1);
}
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(WHITE);
display.setCursor(0, 0);
display.println("Initializing...");
display.display();

sensors.begin();

if (!pox.begin()) {
    Serial.println("MAX30100 failed to initialize!");
    display.clearDisplay();
    display.println("HR Sensor Failed!");
    display.display();
    while (1);
}
pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);

WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
}

```

```

display.clearDisplay();
display.setCursor(0, 10);
display.println("WiFi Connected!");
display.display();
delay(1000);

lastMedicineTime = millis();
}

void loop() {
    pox.update();
    float heartRate = pox.getHeartRate();
    int ecgValue = analogRead(ECG_PIN);

    sensors.requestTemperatures();
    float temperature = sensors.getTempCByIndex(0);

    String hrStatus = getHeartRateStatus(heartRate);
    String timeLeft = getTimeLeft(lastMedicineTime, reminderInterval);

    if (heartRate > 120 || ecgValue > ECG_THRESHOLD) {
        digitalWrite(LED_PIN, HIGH);
        digitalWrite(BUZZER_PIN, HIGH);
    } else {
        digitalWrite(LED_PIN, LOW);
        digitalWrite(BUZZER_PIN, LOW);
    }

    display.clearDisplay();
    display.setCursor(0, 10);
    display.setTextSize(1);
    display.setTextColor(WHITE);
    display.println("Heart Rate: " + String(heartRate) + " BPM");
    display.println("Status: " + hrStatus);
}

```



```

display.println("ECG: " + String(ecgValue));
display.println("Temp: " + String(temperature) + " C");
display.println("Med in: " + timeLeft);

if (millis() - lastMedicineTime >= reminderInterval) {
    display.setCursor(0, 50);
    display.println("Take Medicine Now!");
    lastMedicineTime = millis();
}

display.display();

if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
    String url = server + "?api_key=" + apiKey +
        "&field1=" + String(ecgValue) +
        "&field2=" + String(heartRate);

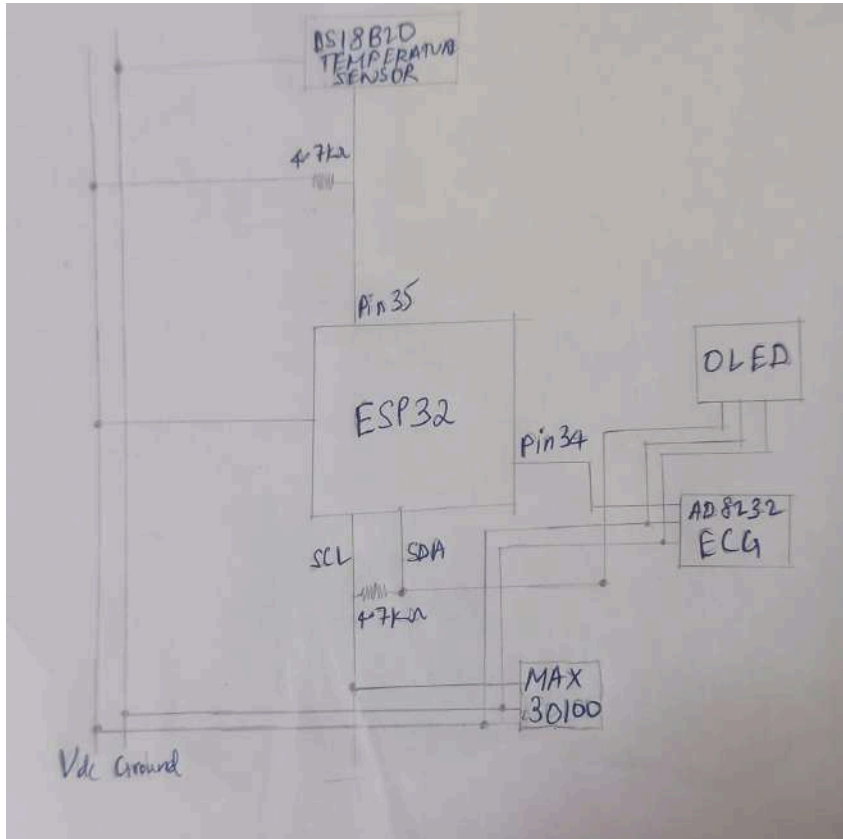
    http.begin(url);
    int httpResponseCode = http.GET();
    http.end();

    Serial.print("ThingSpeak Response: ");
    Serial.println(httpResponseCode);
}

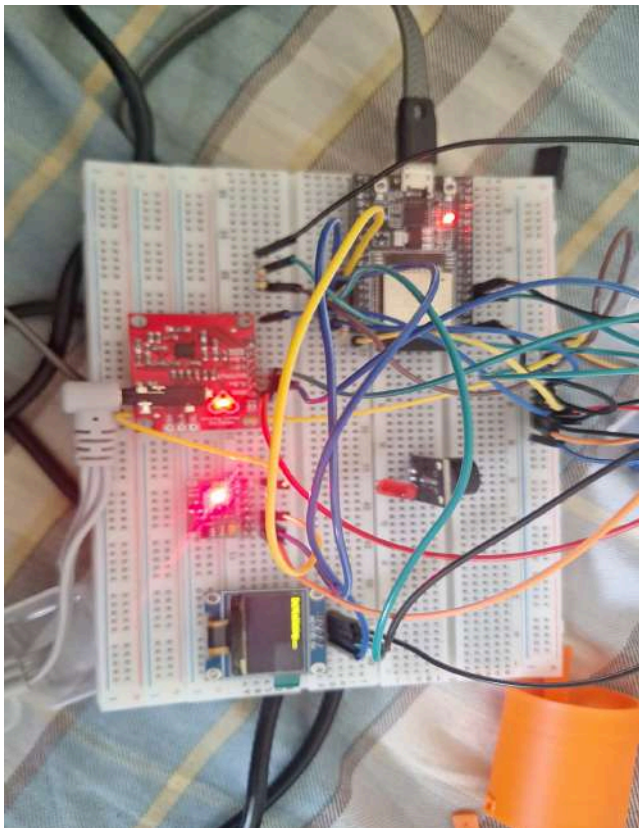
delay(1000);
}

```

2.3 System Block Diagram



2.4 Hardware Picture



3. RESULTS & DISCUSSIONS

The IoT-based healthcare monitoring system developed effectively captures and analyzes biometric data in real-time, facilitating ongoing patient observation and remote access. By incorporating an ESP32 microcontroller along with various biomedical sensors, the system enables real-time monitoring of heart rate, ECG signals, and body temperature while providing timely medication reminders and alerts for emergencies. The data gathered is showcased on an OLED display and sent to a cloud-based platform for oversight by healthcare professionals from a distance. This implementation greatly improves patient safety and access to medical services.

Key Results and Observations

❖ Monitoring Heart Rate and Oxygen Saturation:

- The pulse oximeter effectively records heart rate and oxygen saturation levels in real-time.
- The system classifies heart rate status (normal, low, or high) and activates alerts for irregular readings.

Discussion: Ongoing monitoring of heart rate facilitates the early identification of cardiovascular issues. The system's capability to initiate alerts ensures timely medical intervention, minimizing risks tied to delayed responses. However, factors like sensor positioning and motion disturbances can occasionally impact reading accuracy.

❖ ECG Signal Acquisition:

- The ECG module successfully captures the electrical activity of the heart, facilitating the early identification of abnormalities.
- The analog signal is processed and sent for further examination by healthcare providers.
-

Discussion: The ECG system offers crucial information for identifying arrhythmias and various heart disorders. Nevertheless, the ECG module's susceptibility to noise and movement may necessitate additional filtering techniques to improve precision in practical applications.

❖ Body Temperature Monitoring:

- The temperature sensor constantly tracks body temperature, identifying possible fevers or infections.
- The values are shown on the OLED display and sent for remote monitoring.

Discussion: The incorporation of body temperature tracking facilitates the prompt detection of fever, which is a critical indicator of numerous medical issues.

Environmental conditions may impact the precision of temperature measurements, making it essential to correctly position and calibrate the sensor.

❖ Medication Reminder System:

- A countdown timer is shown to alert patients about their upcoming medication times.
- An alert activates when it's time for the next dose, promoting adherence to medication schedules.

Discussion: The reminder system contributes to better patient compliance with prescribed medication regimens, lowering the likelihood of missed doses. Nonetheless, its effectiveness relies on how patients respond to alerts, which could be improved by incorporating additional alert methods like mobile notifications.

❖ Emergency Alert System:

- When vital health metrics go beyond acceptable limits, an alarm system (including a buzzer and LED) is triggered to alert the user.
- This capability offers real-time feedback, facilitating prompt medical action.

Discussion: The emergency alert system serves as a crucial tool for saving lives by delivering immediate alerts. Nevertheless, false alarms resulting from brief variations in sensor data might need additional optimization to reduce avoidable activations.

❖ Cloud-Based IoT Monitoring:

- All health information is effectively sent to a cloud platform (ThingSpeak) for remote viewing.
- This enables healthcare providers and caregivers to monitor patient health patterns over a period of time.

Discussion: Accessing health information remotely increases the effectiveness of medical oversight, decreasing the necessity for regular hospital visits. Nonetheless, dependence on Wi-Fi connectivity might restrict its operation in regions with inadequate network coverage. Alternative communication methods such as LoRa or GSM could enhance accessibility.

The integration of IoT in healthcare monitoring significantly improves patient supervision by enabling continuous tracking and remote access to vital health data. This system overcomes the limitations of manual health monitoring by providing real-time feedback and automated alerts, ensuring that critical conditions are identified early. While the system effectively captures and processes biometric data, its accuracy can be influenced by external factors such as motion artifacts, sensor placement, and network connectivity. Future improvements, including AI-based health trend analysis, multi-sensor fusion for enhanced accuracy, and alternative connectivity options, could further optimize the system's performance and reliability.

4. CONCLUSION

The healthcare monitoring system utilizing IoT technology effectively allows for the real-time observation of vital signs, enhancing both patient care and the early detection of diseases. By incorporating the ESP32 alongside biomedical sensors, it persistently tracks heart rate, ECG, and body temperature while also offering reminders for medications and alerts for emergencies. The system guarantees remote access through the transmission of data to the cloud, minimizing the necessity for regular hospital visits. Although there are drawbacks such as potential inaccuracies in sensor readings and connectivity challenges, its cost-effectiveness and scalability present a valuable solution for contemporary healthcare. Prospective developments in AI, mobile connectivity, and the expansion of sensor functionalities could further improve its efficiency, thereby making remote monitoring of patients more accessible and dependable.

Key Takeaways:

- Facilitates ongoing, real-time observation of essential health indicators.
- Offers remote access to patient information through cloud storage.
- Incorporates alert systems for unusual vital signs and medication notifications.
- Lowers the frequency of hospital visits, enhancing healthcare accessibility.
- Cost-effective and scalable for wider application.
- Future enhancements in AI and integration of sensors can improve its functionalities.

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